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Preference-Based Assessments

Stability of Danish Population Health Preferences Over Time

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ABSTRACT

Objectives: This study tested the stability of Danish population health preferences by comparing data collected in 2024 with data from the 2019 Danish EQ-5D-5L valuation study. This period included the COVID-19 pandemic and the war in Ukraine.

Methods: Data were collected through (1) 100 virtual interviews following the standardized EQ-VT protocol generating composite time trade-off (cTTO) and discrete choice experiment (DCE) data as in the 2019 study, and (2) an online survey of 1000 respondents generating DCE data. Both samples were matched to the 2019 study. Mean cTTO values were compared between 2019 and 2024. Mixed logit models were estimated for the 2019 and 2024 DCE data, and dimension order was compared. The DCE values were anchored on the quality-adjusted life-years scale using the respective cTTO data and the equivalent states from 2019 and subsequently compared using Bland-Altman plots.

Results: In 2024, 100 EQ-VT interviews were successfully completed, and 1003 respondents had completed the online survey passing quality control. Mean cTTO values did not differ significantly between 2019 and 2024 for the 10 health states valued, although the overall mean was slightly higher in 2024. In the DCE, dimension order was the same in 2019 and 2024; however, the relative importance of mobility and self-care was higher in 2024 at the expense of pain/discomfort and anxiety/depression. Bland-Altman plots showed higher values for severe states and lower values for moderate states in 2024.

Conclusions: Danish population health preferences appeared stable over the 5-year period, providing evidence for the shelf-life of EQ-5D-5L value sets.

Keywords: EQ-5D-5L, stability of health preferences, value sets.

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Highlights

- Limited knowledge exists on the stability of population health preferences over time. To our knowledge, this study is the first of its kind to test the stability of health preferences in the Danish population following the standardized international protocol for EQ-5D-5L valuation studies combined with an online discrete choice experiment survey.
- This study found Danish population health preferences to be robust over a 5-year period that included societal changes related to a pandemic and a nearby war.
- The study provides evidence for the stability of EQ-5D-5L value sets over time.

Introduction

Societal preferences for health are increasingly used around the world to support the prioritization of scarce healthcare resources. The EQ-5D instruments are the most widely used,[1] each measuring health-related quality of life on 5 dimensions—mobility (MO), self-care (SC), usual activities (UA), pain/discomfort (PD), and anxiety/depression (AD)[2,3]—with associated value sets representing societal preferences. The use of value sets in resource allocation underlines the importance of these accurately representing the average societal preferences for health. However, there is only limited empirical evidence on the stability of health preferences over time, and it is not known which factors might warrant the re-estimation of societal value sets or how to determine their expected "shelf-life."[4,5] It is possible, for example, that societal changes or challenges can alter people's attitudes to health or their willingness to trade off length of life for quality of life.[4,5]

Two societal events in recent years are the COVID-19 pandemic and the war in Ukraine. The COVID-19 lockdown affected the general public's day-to-day living by limiting social gatherings and leisure activities—typically belonging under the EQ-5D dimension UA. Another EQ-5D dimension potentially affected by the pandemic is AD because several studies have demonstrated a general worsening of the mental health of affected populations due to social isolation and concerns about one's health and the welfare of one's relatives.[6-8]

Although Denmark did relatively well through the pandemic compared with other countries,[9,10] studies have demonstrated a negative impact on the population's psychological well-being.[11-13] The war in Ukraine has similarly been linked to the deterioration of symptoms in Danish patients with mental disorders despite them being physically distanced from the war.[14] Whether such societal challenges affect the average health preferences of the population is currently unknown. However, it could be speculated that the challenges imposed by these societal events may result in a greater emphasis on UA and AD because these dimensions cover areas where the population has experienced particularly large impacts.

The Danish EQ-5D-5L valuation study was finalized in 2019, immediately before the COVID-19 pandemic, and provides a unique opportunity to assess the stability of health preferences over a 5-year period that included societal challenges related to the pandemic and the war in Ukraine.[15] The aim of this study in

2024 was thus to evaluate the stability of Danish population health preferences through (1) a smaller replication of the 2019 study using the standardized EQ-VT protocol in 100 virtual interviews to generate composite time trade-off (cTTO) and discrete choice experiment (DCE) valuation data and (2) an online survey of 1000 respondents to generate DCE valuation data. The 2019 and 2024 data were compared on mean health state values, the relative importance of the EQ-5D-5L dimensions, and the ordering of dimensions.

Methods

Data for this study were collected through a combination of virtual EQ-VT valuation interviews and an online DCE survey. The data were compared with data from the Danish EQ-5D-5L valuation study that was conducted in 2019 and is reported in detail elsewhere.[15] An overview of the main characteristics of the 2019 and 2024 study designs is provided in Table 1.[15,16] The reporting of this study follows the CREATE checklist in cases in which it is appropriate, and an overview of this is provided in Appendix Material 1 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.05.014).[17]

The EQ-VT Valuation Interview

The valuation interview followed the EQ-VT 2.1 protocol[18] used in the 2019 study, and the Danish version of the EQ-VT software was created by Maths in Health using the original 2019 translations. The interviews were conducted online via Zoom as computer-assisted personal interviews. Each interview consisted of the following items: (1) background questions on age, gender, experience with serious illness, and highest completed education, (2) self-reported health using the EQ-5D-5L questionnaire, (3) cTTO instructions and an example of a cTTO task (wheelchair example), (4) 3 practice cTTO tasks (mild, severe, and difficult to imagine), (5) 10 cTTO valuations of EQ-5D-5L health states, (6) cTTO feedback module for respondents to indicate any health states not ranked appropriately and cTTO debriefing, (7) DCE instructions, (8) DCE valuation of 12 pairs of EQ-5D-5L health states, (9) DCE debriefing, and, finally, (10) additional sociodemographic questions identical to those in 2019, along with new questions on the impact of COVID-19 and the war in Ukraine on the respondent's daily life and well-being.

Preference elicitation techniques in the valuation interview

cTTO and DCE were used as complementary methods for eliciting health state preferences.[18] In cTTO, conventional TTO was used to value health states considered better than dead, whereas lead-time TTO was used to value health states considered worse than dead. The respondent was first presented with a choice between living 10 years in perfect health or 10 years in the EQ-5D-5L health state being valued. If the respondent was indifferent between the 2, the EQ-5D-5L health state was given the value 1; if not, the respondent could trade off time in perfect health until reaching the point of indifference. If the respondent was indifferent between immediate death (ie, trading off all 10 years in perfect health) and living 10 years in the EQ-5D-5L health state, the EQ-5D-5L health state was given the value 0. If the respondent preferred immediate death over living 10 years in the EQ-5D-5L health state, the respondent was given an additional 10 years in perfect health compared with 10 years in perfect health followed by 10 years in the EQ-5D-5L health state (ie, switching to lead-time TTO). Again, the respondent was asked to trade off time in perfect health until the point of indifference was reached. If the respondent traded off all 10 years in perfect health to avoid living 10 years in perfect health followed by 10 years in the EQ-5D-5L health state, the EQ-5D-5L health state was given the value −1.

In the DCE tasks, the respondent was asked to indicate which of 2 EQ-5D-5L health states presented was preferred. The duration of time spent living in the 2 health states was not specified.

Health states valued in the valuation interview

The standard EQ-VT task involves the valuation of 86 health states divided into blocks of 10. In the current virtual EQ-VT interviews, we wished to achieve sufficiently powered direct comparisons with valuations from the 2019 study with a smaller number of respondents; thus, only 1 block of 10 health states was valued by all respondents. This block consisted of 1 mild EQ-5D-5L health state (11112), 8 "moderate" EQ-5D-5L health states (14113, 15151, 21315, 24443, 31524, 43315, 52431, and 54153), and the most severe EQ-5D-5L health state (55555). The order of the 10 health states was randomized.

In the DCE, each respondent completed 12 pairs. A total of 20 blocks of 12 pairs were valued, totaling 240 pairs of health states,

Table 1. Characteristics of the 2019 and 2024 study designs.

Data set	Time point	Type of interview	Design
cTTO data set from 2019*	October 2018-November 2019	Personal assisted in-person interview	Standard EQ-VT design: 86 health states divided into blocks of 10
DCE data set from 2019*	October 2018-November 2019	Personal assisted in-person interview	Standard EQ-VT design: 28 blocks of 7 pairs, totaling 196 pairs of health states
cTTO data set from 2024	December 2023-April 2024	Personal assisted virtual interview	10 health states of 1 block
DCE data set from 2024	December 2023-April 2024	Personal assisted virtual interview	Bailey et al[16] design: 20 blocks of 12 pairs, totaling 240 pairs of health states
DCE online survey data set from 2024	April 2024	Unassisted online survey	Standard EQ-VT design: 28 blocks of 7 pairs, totaling 196 pairs of health states + Bailey et al[16] design: 20 blocks of 12 pairs, totaling 240 pairs of health states

cTTO indicates composite time trade-off; DCE, discrete choice experiment; EQ-VT, EuroQol Valuation Technology.
*Jensen et al.[15]

using the same design as used by Bailey et al,[16] for which the design generation process is described in Jonker and Roudijk.[19] Both the order of the 12 pairs and the left-right positioning were randomized.

Data quality

One skilled interviewer, who was also an interviewer in the 2019 study, conducted all the virtual EQ-VT interviews. The interviewer underwent a “brush-up” using the training material, interview guide, and software instructions from the 2019 study and then carried out 5 pilot interviews. Once the main data collection began, data quality was continuously monitored through quality reports created approximately after every tenth interview using the EQ-VT quality control tool.[20] An interview was flagged as potentially low quality if any of 4 cTTO indicators were observed: (1) the interviewer did not explain “worse than dead” during the wheelchair example, (2) less than 3 minutes were spent on the wheelchair example, (3) logical inconsistency, ie, health state 55555 valued at least 0.5 higher than the lowest ranked health state, and (4) less than 5 minutes were spent on the 10 cTTO tasks.[18,20]

The Online DCE Survey

The online DCE survey comprised the following items: (1) background questions on age, gender, experience of serious illness, and highest completed education (2) self-reported health using the EQ-5D-5L questionnaire, (3) DCE instructions, (4) DCE valuation of 19 pairs of EQ-5D-5L health states, (5) DCE debriefing, and (6) additional sociodemographic questions and questions on the impact of COVID-19 and the war in Ukraine on the respondent’s daily life and well-being. The survey was conducted using Math’s in Health’s LimeSurvey platform.

Health states valued for the online DCE

The online DCE survey aligned with the DCE valuation of health states conducted in the valuation interview. Hence, the respondent was presented with 2 EQ-5D-5L health states and asked to indicate which was preferred. Information on the duration of living in the health states was not provided.

Each respondent completed a total of 19 paired comparisons of EQ-5D-5L health states; a block of 7 pairs following the same design as in the 2019 study to ensure compatibility,[15] henceforth called the “standard EQ-VT design,” and an additional block of 12 pairs from the design used by Bailey et al,[16] for which the design generation process is described in Jonker and Roudijk,[19] henceforth called the Bailey et al[16] design. This new design has several advantages, ie, no mild-mild comparisons, more choice tasks per respondent, and allows for the estimation of mixed logit models and overlap on 2 dimensions. The 7-pair design had 28 blocks, totaling 196 different pairs, whereas the new 12-pair design consisted of 20 blocks, totaling 240 different pairs. The paired comparisons were fully randomized to avoid order effects between the designs.

Data quality

If a respondent spent under 160 seconds on the DCE tasks, their data were considered low quality and were excluded from analysis. This is in line with EQ-5D-Y-3L valuation studies that used 150 seconds on 15 DCE tasks as a threshold for indicating low-quality data.[21,22]

Potential bots (ie, software programs simulating human activity, such as completing a survey) were identified using Google’s reCAPTCHAv3 software incorporated in Maths in Health’s LimeSurvey platform. Based on respondent interaction patterns at 3 stages during the survey, each respondent was assigned a probability of being human. Respondents scoring under 0.5 were considered bots, and their data were excluded from the analysis. Additionally, an open-ended question was added to the survey to enable further identification of bots. Repetitive or incoherent responses were considered indicative of a bot, and the respondent’s data were excluded from further analyses.

Participant Recruitment

Power calculations of previous studies showed that to achieve adequate precision in the measurement of the mean value for a health state, 100 responses per health state were required.[23] Therefore, we aimed for a minimum of 100 high-quality cTTO interviews for analyses. For the DCE, we aimed to collect a sample similar to that collected in the 2019 study. Assuming that dropout due to low-quality data would be comparable to other studies, such as the Dutch and Slovenian EQ-5D-Y-3L valuation studies, approximately 1200 respondents were needed for the online DCE survey to achieve a minimum of 1000 respondents.[22,24] The samples were recruited to match the national sample in the 2019 study regarding age (>18 years), gender, marital status, level of education, and geographic region. A market research company, Norstat, was used to recruit the sample from their panel of respondents. To encourage participation, respondents were offered points that could be used in a web shop hosted by Norstat. Similar to that in the 2019 study, respondents invited to participate in interviews were screened for participation in a health survey within the last 6 months to avoid respondents with strong interview expertise.

Ethics

The study was registered with the Danish Data Protection Agency (case number 2023-068-04169), alongside the 2019 study. Because this study comprised an interview study and a health science questionnaire survey, it did not require approval from the Danish National Committee on Health Research Ethics as per section 14(2) of the Danish Act on Commitees.[25] Information on data anonymity, use, and storage was provided for all respondents in the EQ-VT software.

Statistical Analyses

cTTO data

In this study, cTTO data were collected from a single block of cTTO data, which was also included in the 2019 study.[15] Using two-sample student’s *t* tests, we investigated whether the means of the currently collected responses were significantly different from those collected in the 2019 study. These student’s *t* tests were conducted both between states and over the whole sample. Bonferroni correction was applied when testing for differences between the cTTO responses per state to account for repeated tests of hypotheses.

DCE data

The DCE data were analyzed using mixed logit models. In each of these models, we used a 20-parameter model specification, in which the structural part of the estimated utility function is described as follows:

$$\begin{aligned}U_{ij} = {} & \beta_{i1}MO2_{ij}+\beta_{i2}MO3_{ij}+\beta_{i3}MO4_{ij}+\beta_{i4}MO5_{ij}+\beta_{i5}SC2_{ij}+\beta_{i6}SC3_{ij}\\ & +\beta_{i7}SC4_{ij}+\beta_{i8}SC5_{ij}+\beta_{i9}UA2_{ij}+\beta_{i10}UA3_{ij}+\beta_{i11}UA4_{ij}+\beta_{i12}UA5_{ij}\\ & +\beta_{i13}PD2_{ij}+\beta_{i14}PD3_{ij}+\beta_{i15}PD4_{ij}+\beta_{i16}PD5_{ij}+\beta_{i17}AD2_{ij}\\ & +\beta_{i18}AD3_{ij}+\beta_{i19}AD4_{ij}+\beta_{i20}AD5_{ij}\end{aligned}$$

Here, U_{ij} is the utility associated to state j for respondent i, which is described by a set of coefficients β_{i1} to β_{i20} that represent the utility weights associated with having health problems on any of the EQ-5D-5L dimensions, represented by dummy variables $MO2_{ij}$ to $AD5_{ij}$. These dummy variables equal 1 if the respective level of problems on the respective dimension is present and 0 otherwise. For example, $UA3_{ij}$ equals 1 if state j describes a health state with usual activities at level 3, and 0 otherwise. Incremental dummy coding was used, which means that the estimated beta parameters represent the change in utility for moving from one level to another, eg, β_{i20} is the disutility for moving from AD level 4 to AD level 5.

Because each respondent in this study completed 19 choice pairs (a block of 7 pairs from the standard EQ-VT design and a block of 12 pairs as used in Bailey et al[16]), 5 different models were estimated; (1) a model analyzing the data collected in the 2019 study, (2) a model analyzing just the data from the standard 7-pair EQ-VT blocks from this study, (3) a model analyzing just the data from the 12-pair Bailey et al[16] blocks from this study, (4) the pooled data of 19 pairs per respondent (from both the standard EQ-VT and Bailey et al[16] design) collected in this study,[15] and (5) a garbage-class mixed logit model estimated on the complete set of newly collected data to identify and correct for random response behavior.[26] The model estimates were rescaled using the cTTO data collected in the respective studies, by calculating the mean cTTO response for each of the 10 health states included in this study for both data sets and regressing them onto their respective DCE estimates. The resulting sets of scores, anchored on the 0 (dead) to 1 (full health) quality-adjusted life-year scale, were compared using scatterplots and Bland-Altman plots, and the mean absolute deviation between the estimates was calculated. These analyses were all conducted in Stata version 18.0.

To assess whether there were any significant differences in preferences between the 2019 data and the 2024 data, we estimated a model similar to the one estimated in Jonker et al.[27] We define a conditional logit utility function as follows:

$$U_{tj} = \begin{cases} (\beta_j X_{tj}) + \varepsilon_{tj} \text{ if } t \in T_1 \\ \gamma\theta_d(\beta_j X_{tj}) + \varepsilon_{tj} \text{ if } t \in T_2 \end{cases}$$

Here, U_{tj} is the utility assigned to health state j of choice task t, for which a different type of utility is specified depending on whether t is part of the 2019 sample (T_1) or part of the 2024 sample (T_2).

For both samples, $(\beta_j X_{tj})$ is the inner product of the vector of 20 dummy variables (X_{tj}) representing the presence of certain health problems in the health states being valued, described by the EQ-5D-5L, and a vector of their respective utility decrements β_j. For those choices that were collected in 2024 ($t \in T_2$), we also estimate γ and θ_d, in which γ represents a scale parameter indicating whether there are any differences in the consistency of choices between the 2019 data and 2024 data. θ_d represents a vector of 5 scale parameters that indicate any differences in scale between the 2 samples for each of the 5 EQ-5D-5L dimensions d. The set of 5 parameters was constrained to have an overall mean of 1, which ensures that the scale of the beta parameters is not affected by differences in choice consistency between the 2 samples.[27] An estimated θ parameter larger than 1 then indicates a higher importance for a certain dimension compared with the other sample, and smaller than 1 indicates the opposite. OpenBUGS, a Bayesian estimation software, was used to estimate the model, and informative priors were assumed for each of the estimated parameters. For β_j, multivariate normal distributions with means of 0 and a small standard deviation were assumed.

Table 2. Characteristics of the EQ-VT interview sample and the online DCE survey compared with the 2019 sample from the Danish EQ-5D-5L valuation study.

Characteristics	2024 Study sample		2019 Study sample*
	Interview (*n* = 100)	DCE survey (*n* = 1003)	(*n* = 1014)
Gender			
Female	51 %	51 %	52 %
Male	49 %	49 %	48 %
Other	0 %	0 %	0 %
Age groups (years)			
18-24	<5 %	<5 %	<5 %
25-34	15 %	13 %	13 %
35-44	12 %	13 %	13 %
45-54	23 %	20 %	20 %
55-64	16 %	19 %	18 %
65-74	22 %	22 %	22 %
75+	9 %	10 %	10 %
Marital status			
Widowed	<5 %	5 %	5 %
Divorced	8 %	7 %	11 %
Married	49 %	50 %	50 %
Unmarried	39 %	38 %	34 %
Highest education			
Secondary school	9 %	6 %	8 %
High school/other	6 %	7 %	7 %
Skilled worker	28 %	28 %	27 %
Short-cycle higher education	15 %	13 %	13 %
Medium-cycle higher education	24 %	28 %	28 %
Long-cycle higher education	18 %	19 %	18 %
Work status			
Employed/self-employed	52 %	53 %	54 %
Unemployed (able to work)	<5 %	<5 %	<5 %
Outside the working force	47 %	44 %	42 %
Annual income			
Under DKK 299 999	35 %	30 %	36 %
DKK 300 000-499 999	44 %	37 %	42 %
Over DKK 500 000	19 %	20 %	15 %
Declined to answer	<5 %	<5 %	<5 %
Did not know	<5 %	11 %	<5 %
Geographical region			
North Denmark Region	11 %	10 %	15 %
Central Denmark Region	25 %	23 %	25 %
Region of Southern Denmark	24 %	22 %	19 %
Capital Region of Denmark	27 %	30 %	28 %
Region Zealand	13 %	15 %	13 %

Note. Some percentages do not sum to 100 % because of rounding.
DCE indicates discrete choice experiment; EQ-VT, EuroQol Valuation Technology.
*Adapted from Jensen et al.[15]

For γ, a lognormal distribution was assumed, whereas for θ_d, a multivariate normal distribution with means 1 and small standard deviations was assumed. Three Markov chains with 50 000 burn-ins were used, and 50 000 subsequent iterations were used to estimate the model parameters. Similar to that in Jonker et al,[27] user-written extensions of OpenBUGS were used to ensure the efficiency of sampling and to constrain the means of the scale parameters to 1. The model code is provided in Appendix Material 2 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.05.014. Regular dummy coding was used to ensure the ease of interpretation of the model.

Three different models were estimated: (1) a model combining just the 2019 data and the data from the blocks of 7 pairs from the original EQ-VT DCE design, (2) a model combining the 2019 data and the data from the 12 pairs as used in Bailey et al,[16] and (3) a model comparing the 2019 data and the whole set of DCE data collected in the 2024 study (19 choice pairs per respondent).

Results

Of the 101 EQ-VT interviews completed between December 2023 and April 2024, 1 interview was dropped because of technical issues, leaving 100 interviews for analyses.

The online DCE survey was conducted in April 2024. Of the 1200 respondents who consented to participate, 1063 completed the entire survey. Of these, 34 were flagged as "speeders" (spending less than 160 seconds on the 19 DCE tasks), and 28 were flagged as potential bots ($n < 5$ were flagged as both "speeders" and potential bots). No further respondents were dropped based on the open-ended question, leaving data from 1003 respondents for analyses.

The Sample

The characteristics of both study samples and the 2019 sample are provided in Table 2[15] (numbers < 5 are not provided to avoid microdata reporting). Both current samples were representative of the 2019 sample with only a small overrepresentation of unmarried respondents in the current samples. The DCE survey sample also had an overrepresentation of respondents who reported they did not know their annual income compared with the interview sample and the 2019 sample.

Comparison of Samples

The means for the 10 health states included in the cTTO design collected in this study and the 2019 study are reported in Appendix Material 3 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.05.014. The overall mean differed significantly between the 2 studies (P = .0245, 2-sample student's t test). However, no significant differences were found for any individual health states, with or without Bonferroni correction (2-sample student's t tests).

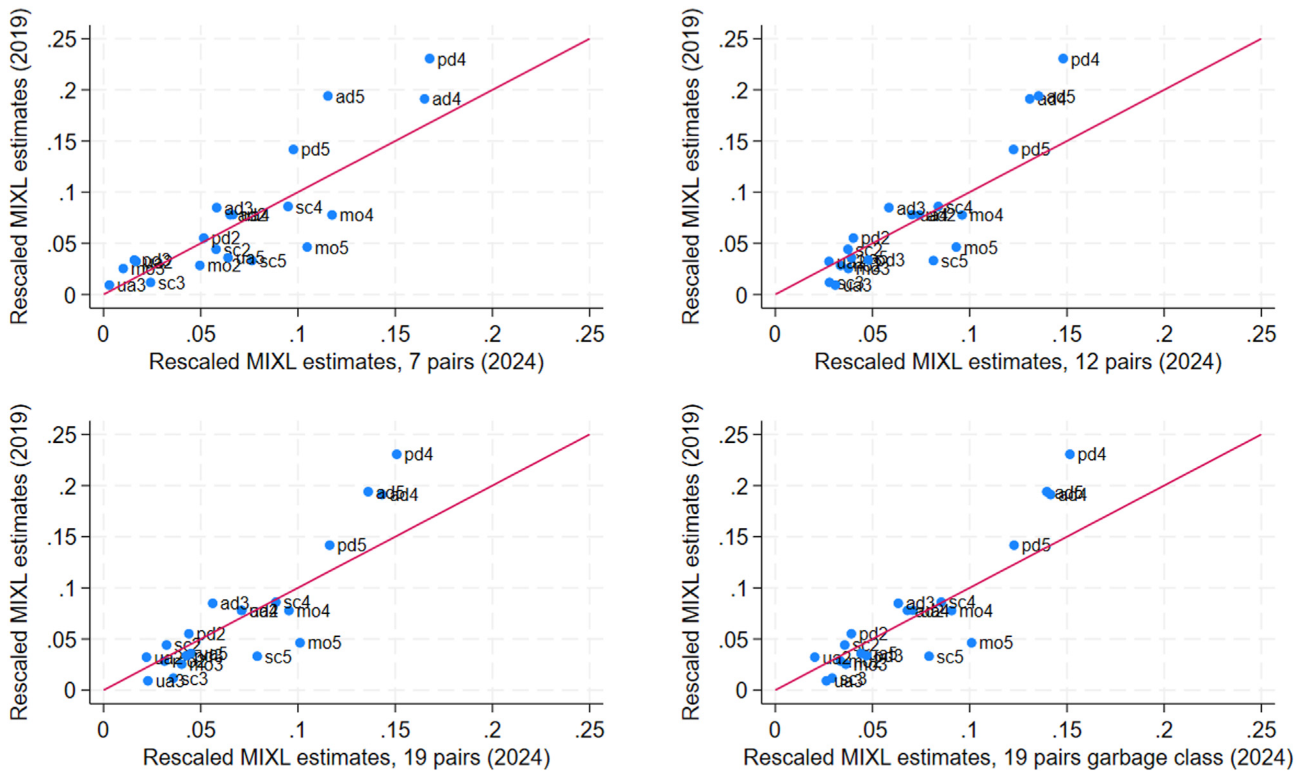
Table 3 reports the rescaled estimates of each of the 5 mixed logit models. Overall, the data collected in the 2019 study show smaller utility decrements for MO and SC compared with any of the models estimated on the currently collected data. For UA, the estimates are roughly similar between all 5 models. The decrements for PD and AD are substantially larger in the model estimated on the 2019 data than in the current data. Figure 1 illustrates the agreement between the coefficients of the model

Table 3. Rescaled mixed logit estimates (incremental dummies).

Dimension level	2019 (MIXL) n = 7098	2024 MIXL 7 pairs n = 7021	2024 MIXL 12 pairs n = 12 036	2024 MIXL (19 pairs) n = 19 057	2024 MIXL (19 pairs) garbage class n = 19 057
mo2	−0.028	−0.050	−0.034	−0.031	−0.034
mo3	−0.025	−0.010	−0.038	−0.040	−0.036
mo4	−0.078	−0.118	−0.096	−0.095	−0.091
mo5	−0.046	−0.105	−0.093	−0.101	−0.101
sc2	−0.044	−0.058	−0.037	−0.032	−0.036
sc3	−0.012	−0.024	−0.028	−0.036	−0.029
sc4	−0.086	−0.095	−0.084	−0.089	−0.085
sc5	−0.033	−0.076	−0.081	−0.079	−0.079
ua2	−0.032	−0.017	−0.028	−0.022	−0.020
ua3	−0.009	−0.003	−0.031	−0.023	−0.026
ua4	−0.078	−0.067	−0.070	−0.071	−0.071
ua5	−0.036	−0.064	−0.039	−0.045	−0.044
pd2	−0.055	−0.052	−0.040	−0.044	−0.039
pd3	−0.034	−0.016	−0.048	−0.043	−0.047
pd4	−0.231	−0.168	−0.148	−0.151	−0.152
pd5	−0.142	−0.098	−0.123	−0.116	−0.123
ad2	−0.078	−0.065	−0.074	−0.071	−0.068
ad3	−0.085	−0.058	−0.058	−0.056	−0.063
ad4	−0.191	−0.165	−0.131	−0.143	−0.142
ad5	−0.194	−0.115	−0.136	−0.136	−0.140

ad indicates anxiety/depression; MIXL, mixed logit; mo, mobility; pd, pain/discomfort; sc, self-care; ua, usual activities.

Figure 1. Comparison of rescaled coefficients for the DCE models. The vertical axis reports estimates from the 2019 data, and the horizontal axis reports estimates for different subsets of the 2024 data. The sample sizes are the same as those reported in Table 3.



ad indicates anxiety/depression; DCE, discrete choice experiment; MIXL, mixed logit; mo, mobility; pd, pain/discomfort; sc, self-care; ua, usual activities.

estimated using the 2019 data and the 4 models using the currently collected data. Figure 2A (scatterplot) and B (Bland-Altman plot) illustrate agreement in the rescaled values for the model estimated using the 2019 data and one comparator model estimated on the current data, showing small differences between the estimated values for milder and moderate states and larger differences for the more severe health states. The raw estimates are presented in Appendix Material 4 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.05.014.

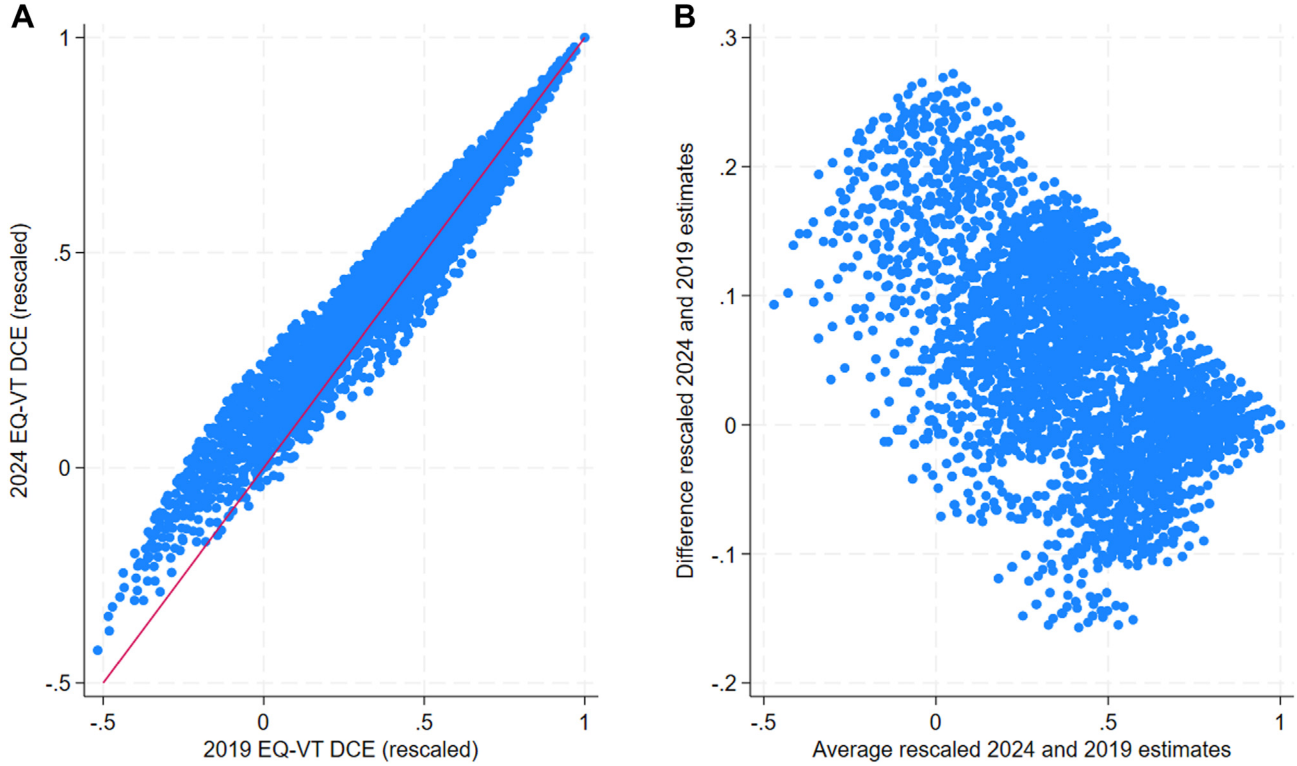
The scale parameter models in Table 4 show that MO and SC are significantly more important in the 2024 data compared with 2019, with mobility being between 36.3% and 30.4% more important. SC was between 31.7% and 20.0% more important in the 2024 sample compared with 2019, whereas the importance of UA did not differ between samples. PD was between 31.7% and 22.8% less important, and AD was between 30.4% and 26.1% less important in the 2024 data compared with 2019. Choice consistency, as indicated by the gamma parameter, was lower when comparing just the 2019 data with the 2024 data collected using 7-pair DCE design, whereas it was higher in the 2024 data when comparing with the 12-pair design. For the full sample, there was no difference in consistency.

Discussion

To our knowledge, this study is the first of its kind to test the stability of population preferences for health following the standardized international protocol for EQ-5D-5L valuation studies.[18] The overall results show that Danish population health preferences appeared stable over a 5-year period that included the COVID-19 pandemic and the war in Ukraine. The cTTO data showed no statistically significant changes when health states were directly compared, although the overall mean in 2024 was slightly higher than that in 2019. For the DCE, the relative importance of the dimensions was altered, for which the 2024 data gave slightly more importance to MO and SC and less importance to AD and PD, but the ordering of dimensions was the same as in 2019.

Few other studies have looked at the stability of preferences. Rupel and Ogorevc found EQ-5D preferences (as EQ-5D-3L) in the Slovenian population to be stable over a 5-year period from 2000 to 2005, although they used different valuation methodologies in their 2 studies and had relatively small sample sizes.[28] National preferences for health could be altered by an aging population or large changes in ethnicity or political learning, as well as changes in living standards or in awareness of mental health issues.[29] Regarding this study, these factors seem unlikely to have been relevant during the 5-year study period. We aimed to replicate the 2019 Danish EQ-5D-5L valuation study as closely as possible to minimize methodological differences. Valuation studies are resource consuming, however, and because we were not aiming for a new Danish EQ-5D-5L value set, we chose to perform a small-scale replica of the 2019 study and couple this with an online DCE survey. The mode of EQ-VT interview differed between the 2019 study and this study because the 2019

Figure 2. (A) Scatter plot showing rescaled DCE estimates 2024 versus 2019 (MIXL 7 pairs). The sample sizes are the same as those in Table 3. (B) Bland-Altman plot showing rescaled DCE estimates 2024 versus 2019 (MIXL 7 pairs). The sample sizes are the same as those in Table 3.



DCE indicates discrete choice experiment; MIXL, mixed logit.

interviews were conducted face to face, and those in this study were online using videoconference. However, Rowen et al[30] and Peasgood et al[31] showed that the mode of interview did not cause significant differences in cTTO valuations. Peasgood et al[31] found that online interviews were preferred over face-to-face interviews, and there were no differences in participant engagement or data quality indicators.[31] Sampling bias is possible, however, because respondents agreeing to participate online may be different to those agreeing to participate face to face.

It was a strength of this study that only 1 interviewer conducted all the interviews and that she also participated in the 2019 study. Although 11 interviewers participated in the 2019 study, we believe that interviewer effects were minimal because data quality and protocol compliance were high.[15]

A limitation of the online DCE survey was the risk of bots acting as respondents. In an attempt to identify and eliminate potential bots, this study used the reCAPTCHAv3 software incorporated in Maths in Health's LimeSurvey platform and an open-ended question that was reviewed manually for all respondents. Furthermore, the market research company Norstat has a strict inclusion procedure for respondents in their panel, for which a personal invitation is needed to join.[32] Combined, the issue of bots in the final sample was believed to be limited.

In this study, respondents completed 19 DCE tasks in the online DCE survey, combining both the standard EQ-VT design and a design used by Bailey et al,[16] whereas the 2019 study only used the standard EQ-VT design. The expanded DCE design was chosen to ensure a robust comparison between the 2019 data collected face to face and the online DCE survey that was expected to provide lower-quality data. In practice, our analysis showed that estimates based on the data from the 2 designs used in this study did not differ substantially on the most important characteristics. When comparing the data collected in 2019 with the data collected in 2024 using the same DCE design, the online collected data show lower consistency. However, the 2024-collected data using the Bailey et al[16] design show a much higher consistency, whereas the combined sample shows similar consistency to the 2019 data, suggesting that the full sample of 2024-collected data may be a viable comparator.

Collecting cTTO data is logistically challenging and expensive, which makes it challenging to test whether health preferences have changed over time using this method. Alternative methods using online-only data collection might be preferable, but values on the 0 to 1 quality-adjusted life-year scale are needed. This disqualifies the use of DCE as currently implemented in the EQ-VT protocol and used in this study. An alternative method could be DCE with a duration attribute, which has been shown to be able to produce similar estimates to cTTO data collections and, in the future, might be an attractive option to test the stability of preferences.[33]

The study findings indicate that the Danish EQ-5D-5L value sets established before the COVID-19 pandemic are still relevant for healthcare decision making. Future research could include a repetition of this study after an additional 5 to 10 years to provide further empirical evidence for the "shelf-life" of population health preferences and, in particular, of EQ-5D-5L value sets.

Table 4. Interaction models estimating differences between the 2019 data and different sets of the 2024 data.

Dimension level	Old data versus 7-pair EQ-VT design data				Old data versus 12-pair EQ-VT design data				Old data versus full sample new data			
	Mean	SD	2.5% perc	97.5% perc	Mean	SD	2.5% perc	97.5% perc	Mean	SD	2.5% perc	97.5% perc
mo2	−0.189	0.040	−0.268	−0.112	−0.143	0.027	−0.198	−0.090	−0.132	0.028	−0.187	−0.078
mo3	−0.248	0.047	−0.342	−0.157	−0.295	0.031	−0.358	−0.235	−0.282	0.033	−0.347	−0.219
mo4	−0.736	0.057	−0.851	−0.626	−0.725	0.048	−0.820	−0.632	−0.707	0.049	−0.804	−0.614
mo5	−1.127	0.071	−1.269	−0.989	−1.125	0.069	−1.259	−0.991	−1.125	0.069	−1.262	−0.991
sc2	−0.241	0.045	−0.330	−0.155	−0.200	0.030	−0.261	−0.142	−0.207	0.031	−0.268	−0.148
sc3	−0.350	0.048	−0.445	−0.257	−0.348	0.033	−0.415	−0.284	−0.362	0.034	−0.430	−0.297
sc4	−0.794	0.059	−0.913	−0.681	−0.739	0.049	−0.837	−0.646	−0.732	0.049	−0.830	−0.638
sc5	−1.124	0.068	−1.261	−0.992	−1.099	0.064	−1.227	−0.976	−1.106	0.065	−1.235	−0.980
ua2	−0.206	0.050	−0.306	−0.109	−0.153	0.034	−0.220	−0.088	−0.149	0.036	−0.220	−0.080
ua3	−0.139	0.056	−0.249	−0.030	−0.273	0.036	−0.347	−0.204	−0.232	0.039	−0.309	−0.158
ua4	−0.674	0.063	−0.801	−0.551	−0.700	0.055	−0.810	−0.595	−0.693	0.055	−0.804	−0.587
ua5	−0.894	0.071	−1.035	−0.758	−0.917	0.066	−1.050	−0.789	−0.909	0.067	−1.042	−0.781
pd2	−0.485	0.058	−0.599	−0.371	−0.369	0.041	−0.449	−0.289	−0.352	0.044	−0.438	−0.267
pd3	−0.549	0.062	−0.671	−0.429	−0.624	0.044	−0.710	−0.539	−0.580	0.047	−0.673	−0.490
pd4	−1.831	0.071	−1.972	−1.693	−1.772	0.062	−1.895	−1.653	−1.766	0.063	−1.891	−1.644
pd5	−2.540	0.085	−2.707	−2.376	−2.574	0.081	−2.734	−2.418	−2.556	0.081	−2.716	−2.399
ad2	−0.528	0.059	−0.647	−0.412	−0.567	0.044	−0.655	−0.481	−0.546	0.045	−0.636	−0.458
ad3	−0.996	0.063	−1.120	−0.873	−1.034	0.049	−1.132	−0.939	−0.990	0.050	−1.089	−0.894
ad4	−2.134	0.079	−2.289	−1.982	−2.066	0.068	−2.202	−1.935	−2.073	0.069	−2.211	−1.939
ad5	−2.997	0.092	−3.177	−2.818	−3.074	0.089	−3.252	−2.905	−3.053	0.090	−3.229	−2.876
gamma	0.858	0.039	0.785	0.937	1.349	0.063	1.229	1.476	0.982	0.040	0.907	1.064
theta mo	1.363	0.075	1.220	1.512	1.304	0.062	1.187	1.433	1.304	0.063	1.187	1.432
theta sc	1.317	0.074	1.178	1.468	1.200	0.059	1.088	1.321	1.270	0.062	1.154	1.395
theta ua	0.899	0.076	0.757	1.054	1.027	0.063	0.911	1.159	0.965	0.061	0.851	1.094
theta pd	0.683	0.035	0.615	0.754	0.772	0.030	0.715	0.831	0.732	0.029	0.677	0.789
theta ad	0.739	0.032	0.676	0.803	0.696	0.024	0.649	0.745	0.729	0.025	0.680	0.779

ad indicates anxiety/depression; mo, mobility; pd, pain/discomfort; perc, percentile; sc, self-care; SD, standard deviation; ua, usual activities.

Conclusions

This study found that Danish EQ-5D-5L population health preferences measured using time trade-off and DCE appeared robust over a 5-year time period, or at least that any changes in preferences were transient. This time period included societal challenges related to the COVID-19 pandemic and the war in Ukraine.

Author Disclosures

Author disclosure forms can be accessed below in the Supplemental Material section. Views expressed by the authors in the publication do not necessarily reflect the views of the EuroQol Foundation.

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